

yone_calculations

February 11, 2025

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[59]: from pandas import DataFrame
      from dataframe_image import export
      from IPython.display import Image, display
      from os import path
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0.1 Yone – Blade of the Ruined King vs Yun Tal Wildarrows

For brevity, we abbreviate these items as BRK and YT, as well as Immortal Shieldbow -> SB, Infinity Edge -> IE. In order to contrast the value between BRK and YT, it is valuable to assess various metrics at different points in the game, and it is helpful to abstract some of the computation away

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[60]: class Enemy:
      def __init__(self, HP, armor, MR):
          self.HP = HP
          self.armor = armor
          self.MR = MR

      def take_phys_dmg(self, damage):
          return damage * (100 / (100 + self.armor))

      def take_mag_dmg(self, damage):
          return damage * (100 / (100 + self.MR))

      class Item:
          def __init__(self, price, AD, AS, crit_chance, crit_damage, on_hit_effect):
              self.price = price
              self.AD = AD
              self.AS = AS
              self.crit_chance = crit_chance
              self.crit_damage = crit_damage
              self.on_hit_effect = on_hit_effect

      def brk_on_hit(level, target):
          # it deals 8% current HP damage, so to help it a little bit,
          # we'll assume the targets are 60% HP
          return target.HP * 0.6 * 0.08
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BRK = Item(3200, 40, 25, 0, 0, brk_on_hit)
YT_off_half = Item(3000, 55, 35, 12.5, 0, None) # half stacked YT without
↳passive
YT_on_half = Item(3000, 55, 65, 12.5, 0, None) # half stacked YT with passive
YT_off = Item(3000, 55, 35, 25, 0, None)
YT_on = Item(3000, 55, 65, 25, 0, None)
IE = Item(3600, 70, 0, 25, 40, None)
SB = Item(3000, 55, 0, 25, 0, None)

class Yone:
    def __init__(self, level, alacrity_stacks, items):
        self.level = level
        # from base stats + levels
        self.AD = 60 + 2 * (level - 1)
        self.AS = 3.5 * (level - 1)
        # from runes (stat shards, alacrity, lethal_tempo)
        self.AD += 5.4
        self.AS += 10
        self.AS += 3 + 1.5 * alacrity_stacks
        self.lethal_tempo = False
        self.on_hit_effects = []
        def lethal_tempo_bolt(level, target):
            if self.lethal_tempo:
                base_damage = 9 + (17/21) * (level - 1)
                return base_damage * (self.AS/100 + 1)
            else:
                return 0
        self.on_hit_effects.append(lethal_tempo_bolt)
        # from items (auto-include doran's blade and berserker's greaves)
        self.AD += 10
        self.AS += 25
        self.price = 1550
        self.crit_chance = 0
        self.crit_damage = 175 * 0.9
        for item in items:
            self.price += item.price
            self.AD += item.AD
            self.AS += item.AS
            self.crit_chance += item.crit_chance * 2
            self.crit_damage += item.crit_damage * 0.9
            if item.on_hit_effect:
                self.on_hit_effects.append(item.on_hit_effect)

    def toggle_lethal_tempo(self):
        if self.lethal_tempo:
            self.AS -= 36
        else:

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        self.AS += 36
        self.lethal_tempo = not self.lethal_tempo

    def apply_on_hits(self, target):
        on_hit_dmg = 0
        for on_hit_effect in self.on_hit_effects:
            on_hit_dmg += on_hit_effect(self.level, target)
        # on hits are purely physical dmg
        return target.take_phys_dmg(on_hit_dmg)

    def calc_AA_dmg(self, target):
        basic_dmg = self.AD * (1 + (self.crit_chance/100) * (self.crit_damage/
↪100 - 1))
        # 75% of the damage from AAs is physical, 25% is magic
        total = target.take_phys_dmg(basic_dmg * 0.75) + target.
↪take_mag_dmg(basic_dmg * 0.25)
        # on hits are entirely physical
        total += self.apply_on_hits(target)
        return total

    def calc_APS(self):
        # attacks per second
        AS_ratio = 0.625
        return AS_ratio * (self.AS/100 + 1)

    def calc_AA_DPS(self, target):
        return self.calc_AA_dmg(target) * self.calc_APS()

    def calc_Q_dmg(self, target):
        # determine Q rank
        match self.level:
            case 1 | 2 | 3:
                rank = 1
            case 4:
                rank = 2
            case 5 | 6:
                rank = 3
            case 7 | 8:
                rank = 4
            case _:
                rank = 5
        base_damage = 20 + 25 * (rank - 1)
        # the AD scaling component can crit, for 90% damage like his AAs
        AD_scaling = 1.05 * self.AD * (1 + (self.crit_chance/100) * (self.
↪crit_damage/100 - 1))
        # Q is entirely physical damage

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        return target.take_phys_dmg(base_damage + AD_scaling) + self.
↪apply_on_hits(target)

def calc_Q_CD(self):
    CD = 4 * (1 - (0.01 * (self.AS/1.67)))
    if CD < 1.33:
        CD = 1.33
    return CD

def calc_Q_DPS(self, target):
    # cast time shouldn't matter because Q is on CD anyway
    return self.calc_Q_dmg(target) * (1/self.calc_Q_CD())

def calc_W_CD(self):
    CD = 14 * (1 - (0.01 * (self.AS/1.51)))
    if CD < 6:
        CD = 6
    return CD

def calc_total_DPS(self, target):
    # of course, in-game, cast time of Q might interfere with AA rhythm,
    # but the effect should be minimal so we are ignoring it
    return self.calc_AA_DPS(target) + self.calc_Q_DPS(target)

```

Now, let's consider some specific situations you might find yourself in. First, let's consider a 1 item Yone (around ~14 minutes probably, with Doran's Blade + Berserker's Greaves and an half-stacked alacrity – and half-stacked YT if applicable). Let's consider them hitting a tank, who presumably has an armor tank item and defensive boots (although I could not be bothered to implement the Plated Steelcaps damage reduction, so we'll just give them even more armor to compensate), a bruiser champion who presumably has some HP from their first item and possibly defensive boots, and a squishy, who presumably has an offensive item and offensive boots. From some of the LR scrimms I'm watching, at ~14 minutes, the solo lanes tend to be around levels 10-11 at this time, bot laners are level ~9, and supps are level ~7. With this, I'll say that tanks have around 2000 HP with 130 armor and 60 MR, bruisers have around 2000 HP with 85 armor and 55 MR, and squishies have 1500 HP with 65 armor and 45 armor. Of course, this varies greatly depending on the champ and what items the build, and runes like scaling HP, overgrowth, etc. The squishy stats are also a bit leading because a squishy mid will be slightly tankier than the numbers I listed, while a squishy bot laner will be less tanky, and a squishy support will be substantially less tanky. Also, support tanks are closer to bruisers in terms of durability, but tend to have way more resistances and less HP (because of Locket's stat profile and aftershock), which lessens the value of BRK against them. In general, these values are helpful baselines to examine, but do not consider any nuance and therefore can be misleading. As for two items, it's reasonable to assume the Tank now has an armor item and a mixed item (like Despair, even though it's actually bad and getting killed o7, or Jak'Sho), or possibly a full MR item, or maybe an HP item like Fimbul. In any case, we'll consider the "average" of these. As for the bruiser, we'll just assume they pick up another HP item like Cleaver, Sundered Sky, Sterak's, Riftmaker, Rylai's, etc. Then, for the squishy, we just assume another offensive item. At this point in the game, ~22 minutes, we can assume the solo lanes are levels 13-14, the bot laner is level 12-13, and the supps are levels 9-10. Of course, once again, this is

ignoring much of the nuance, but I think it is reasonable to give the tank 3200 HP with 165 armor and 100 MR, the bruiser 3200 HP with 120 armor and 70 MR, and the squishy 2100 HP with 95 armor and 55 MR. Bot laners and supports should actually be much squishier than this, but this is accounting for high level solo laners with items like Liandry or Trinity which give HP.

```
[61]: def create_dataframe(yones, enemy):
    array = []
    for yone in yones:
        subarray = []
        subarray.append(yone.calc_AA_DPS(enemy))
        subarray.append(yone.calc_Q_dmg(enemy))
        subarray.append(yone.calc_Q_CD())
        subarray.append(yone.calc_Q_DPS(enemy))
        subarray.append(yone.calc_W_CD())
        subarray.append(yone.calc_total_DPS(enemy))
        subarray.append(yone.price)
        subarray.append(yone.calc_total_DPS(enemy) / yone.price)
        array.append(subarray)
    row_labels = ["BRK", "YT Off", "YT On"]
    column_labels = ["AA DPS", "Q DMG", "Q CD", "Q DPS", "W CD", "Q + AA DPS", "Q + AA DPS / Gold", "Q + AA DPS / Gold"]
    return DataFrame(array, index=row_labels, columns=column_labels)

def style_and_display_dataframe(df, filename, title):
    styled_df = df.style.background_gradient(cmap="coolwarm", axis=0).format("{:.2f}")
    styled_df.set_caption(title)
    # I have a folder called "figures" to hold these
    full_path = path.join("figures", filename)
    export(styled_df, full_path)
    display(Image(full_path))

# -- Round 1 -- One item --
tank_1 = Enemy(2000, 130, 60)
bruiser_1 = Enemy(2000, 85, 55)
squishy_1 = Enemy(1500, 65, 45)
enemies_1 = [tank_1, bruiser_1, squishy_1]
yone_1_brk = Yone(10, 5, [BRK])
yone_1_yt_off = Yone(10, 5, [YT_off_half])
yone_1_yt_on = Yone(10, 5, [YT_on_half])
yones_1 = [yone_1_brk, yone_1_yt_off, yone_1_yt_on]

# -- Round 2 -- SB second --
tank_2 = Enemy(3200, 165, 100)
bruiser_2 = Enemy(3200, 120, 70)
squishy_2 = Enemy(2100, 95, 55)
enemies_2 = [tank_2, bruiser_2, squishy_2]
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yone_2_brk = Yone(13, 10, [BRK, SB])
yone_2_yt_off = Yone(13, 10, [YT_off, SB])
yone_2_yt_on = Yone(13, 10, [YT_on, SB])
yones_2 = [yone_2_brk, yone_2_yt_off, yone_2_yt_on]

# -- Round 3-1 -- IE second, Tempo Off --
enemies_3 = enemies_2
yone_3_brk = Yone(13, 10, [BRK, IE])
yone_3_yt_off = Yone(13, 10, [YT_off, IE])
yone_3_yt_on = Yone(13, 10, [YT_on, IE])
yones_3 = [yone_3_brk, yone_3_yt_off, yone_3_yt_on]

all_yones = [yones_1, yones_2, yones_3]
all_enemies = [enemies_1, enemies_2, enemies_3]

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[62]: for i in range(3):
        yones = all_yones[i]
        enemies = all_enemies[i]
        title_component_1 = ["One item", "SB second", "IE second"]
        for j in range(3):
            title_component_2 = ["vs Tank", "vs Bruiser", "vs Squishy"]
            df_1 = create_dataframe(yones, enemies[j])
            filename_1 = f"yone_figure_r{i+1}_{j+1}_1.png"
            base_title = f"{title_component_1[i]} {title_component_2[j]}"
            title_1 = base_title + ", Tempo off"
            style_and_display_dataframe(df_1, filename_1, title_1)
            for yone in yones:
                yone.toggle_lethal_tempo()
            df_2 = create_dataframe(yones, enemies[j])
            filename_2 = f"yone_figure_r{i+1}_{j+1}_2.png"
            title_2 = base_title + ", Tempo on"
            style_and_display_dataframe(df_2, filename_2, title_2)
            for yone in yones:
                yone.toggle_lethal_tempo()

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One item vs Tank, Tempo off									
	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold	
BRK	133.93	154.81	1.56	99.44	6.00	233.37	4750.00	0.05	
YT Off	108.48	129.66	1.33	97.49	6.00	205.96	4550.00	0.05	
YT On	123.83	129.66	1.33	97.49	6.00	221.31	4550.00	0.05	

One item vs Tank, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	182.87	171.67	1.33	129.07	6.00	311.94	4750.00	0.07
YT Off	154.11	147.22	1.33	110.69	6.00	264.81	4550.00	0.06
YT On	176.45	149.34	1.33	112.29	6.00	288.74	4550.00	0.06

One item vs Bruiser, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	160.95	192.47	1.56	123.63	6.00	284.58	4750.00	0.06
YT Off	127.45	161.20	1.33	121.20	6.00	248.65	4550.00	0.05
YT On	145.48	161.20	1.33	121.20	6.00	266.69	4550.00	0.06

One item vs Bruiser, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	220.81	213.42	1.33	160.47	6.00	381.27	4750.00	0.08
YT Off	182.93	183.03	1.33	137.62	6.00	320.55	4550.00	0.07
YT On	209.65	185.67	1.33	139.60	6.00	349.25	4550.00	0.08

One item vs Squishy, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	160.68	201.25	1.56	129.27	6.00	289.95	4750.00	0.06
YT Off	141.00	180.74	1.33	135.89	6.00	276.89	4550.00	0.06
YT On	160.95	180.74	1.33	135.89	6.00	296.85	4550.00	0.07

One item vs Squishy, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	224.26	224.75	1.33	168.98	6.00	393.24	4750.00	0.08
YT Off	202.88	205.22	1.33	154.30	6.00	357.18	4550.00	0.08
YT On	232.57	208.18	1.33	156.52	6.00	389.10	4550.00	0.09

SB second vs Tank, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	220.12	202.42	1.33	152.19	6.00	372.31	7750.00	0.05
YT Off	193.44	175.96	1.33	132.30	6.00	325.74	7550.00	0.04
YT On	218.67	175.96	1.33	132.30	6.00	350.97	7550.00	0.05

SB second vs Tank, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	285.06	220.50	1.33	165.79	6.00	450.85	7750.00	0.06
YT Off	254.95	194.75	1.33	146.43	6.00	401.37	7550.00	0.05
YT On	287.62	196.86	1.33	148.02	6.00	435.64	7550.00	0.06

SB second vs Bruiser, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	263.93	243.82	1.33	183.32	6.00	447.26	7750.00	0.06
YT Off	231.34	211.95	1.33	159.36	6.00	390.71	7550.00	0.05
YT On	261.52	211.95	1.33	159.36	6.00	420.88	7550.00	0.06

SB second vs Bruiser, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	341.97	265.60	1.33	199.70	6.00	541.66	7750.00	0.07
YT Off	305.17	234.58	1.33	176.38	6.00	481.55	7550.00	0.06
YT On	344.31	237.13	1.33	178.29	6.00	522.60	7550.00	0.07

SB second vs Squishy, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	258.95	248.00	1.33	186.47	6.00	445.42	7750.00	0.06
YT Off	258.81	239.13	1.33	179.79	6.00	438.60	7550.00	0.06
YT On	292.57	239.13	1.33	179.79	6.00	472.36	7550.00	0.06

SB second vs Squishy, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	340.63	272.57	1.33	204.94	6.00	545.57	7750.00	0.07
YT Off	341.76	264.65	1.33	198.99	6.00	540.75	7550.00	0.07
YT On	385.63	267.53	1.33	201.15	6.00	586.78	7550.00	0.08

IE second vs Tank, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	252.10	225.00	1.33	169.18	6.00	421.27	8350.00	0.05
YT Off	254.68	217.33	1.33	163.41	6.00	418.08	8150.00	0.05
YT On	287.90	217.33	1.33	163.41	6.00	451.30	8150.00	0.06

IE second vs Tank, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	322.28	243.08	1.33	182.77	6.00	505.05	8350.00	0.06
YT Off	325.77	236.11	1.33	177.53	6.00	503.30	8150.00	0.06
YT On	366.43	238.23	1.33	179.12	6.00	545.55	8150.00	0.07

IE second vs Bruiser, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	302.18	271.03	1.33	203.78	6.00	505.96	8350.00	0.06
YT Off	304.58	261.78	1.33	196.83	6.00	501.41	8150.00	0.06
YT On	344.31	261.78	1.33	196.83	6.00	541.14	8150.00	0.07

IE second vs Bruiser, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	386.47	292.80	1.33	220.15	6.00	606.62	8350.00	0.07
YT Off	389.87	284.41	1.33	213.84	6.00	603.72	8150.00	0.07
YT On	438.56	286.96	1.33	215.76	6.00	654.33	8150.00	0.08

IE second vs Squishy, Tempo off

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	301.74	278.70	1.33	209.55	6.00	511.29	8350.00	0.06
YT Off	340.74	295.35	1.33	222.06	6.00	562.81	8150.00	0.07
YT On	385.19	295.35	1.33	222.06	6.00	607.25	8150.00	0.07

IE second vs Squishy, Tempo on

	AA DPS	Q DMG	Q CD	Q DPS	W CD	Q + AA DPS	Gold	Q + AA DPS / Gold
BRK	390.42	303.27	1.33	228.02	6.00	618.44	8350.00	0.07
YT Off	436.52	320.87	1.33	241.26	6.00	677.78	8150.00	0.08
YT On	491.08	323.75	1.33	243.42	6.00	734.50	8150.00	0.09

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[ ]: # jupyter nbconvert --to pdf yone_calculations.ipynb
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